

LAB # 4**DECISION MAKING IN PROGRAMMING USING IF ELSE
STRUCTURE****Lab Tasks**

- 1- Mention the output for the following program:

```
#include<stdio.h>

void main()
{
    int a=100;
    if(a>10)
        printf("Shahid Afridi");
    else if(a>20)
        printf("Shoaib Akhtar");
    else if(a>30)
        printf("Kamran Akmal");
}
```

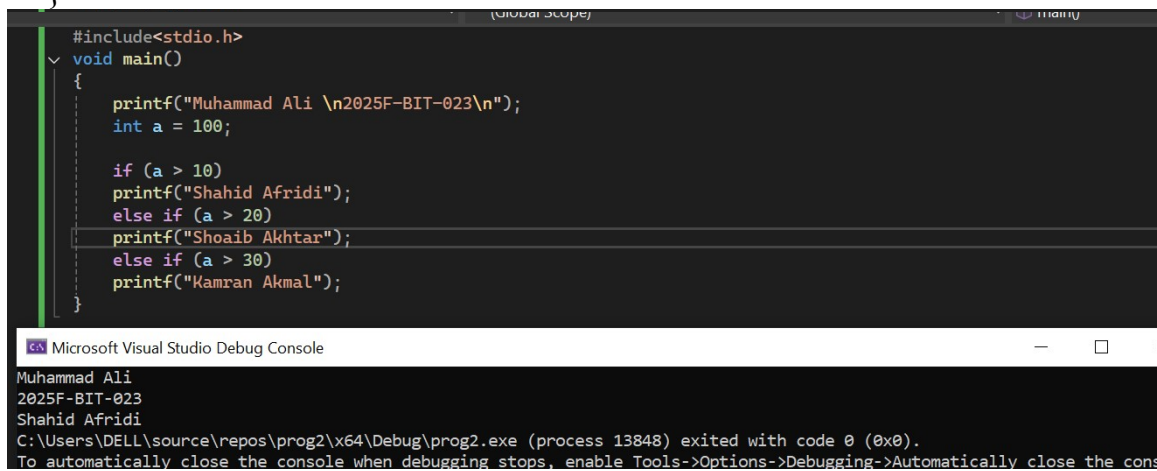
- 2- Write a program that takes a number as input from user and checks whether the Number is even or odd.
Using if-else:
- 3- Write a program that declares and initializes two numbers with your_roll_no and your_friend_roll_no and displays the greater of the two.
- 4- Write a program to read the age of a candidate and determine whether it is eligible for casting his/her own vote.
- 5- Write a program to find the maximum between three numbers by using the if-else statement.

- 6- Write a program to take a value from the user as input marks of five subjects Physics, Chemistry, Biology, Mathematics, and Computer. Calculate percentage and grade according to the following:
- Percentage $\geq 90\%$: Grade A,
 - Percentage $\geq 80\%$: Grade B,
 - Percentage $\geq 70\%$: Grade C,
 - Percentage $\geq 60\%$: Grade D,
 - Percentage $\geq 40\%$: Grade E,
 - Percentage $< 40\%$: Grade F,
- Write this program with the help of if-else statement.

1.Mention the output for the following program:

```
#include<stdio.h>

void main()
{
    int a=100;
    if(a>10)
        printf("Shahid Afridi");
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        printf("Kamran Akmal");
}
```



The screenshot shows the Microsoft Visual Studio IDE. The top pane displays the C code from the previous block. The bottom pane, titled 'Microsoft Visual Studio Debug Console', shows the output of the program. The output consists of three lines: 'Muhammad Ali', '2025F-BIT-023', and 'Shahid Afridi'. Below the output, a message indicates that the process 'C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 13848) exited with code 0 (0x0)'. A final line of text suggests enabling 'Tools->Options->Debugging->Automatically close the console' for better debugging experience.

```
#include<stdio.h>
void main()
{
    printf("Muhammad Ali \n2025F-BIT-023\n");
    int a = 100;

    if (a > 10)
        printf("Shahid Afridi");
    else if (a > 20)
        printf("Shoaib Akhtar");
    else if (a > 30)
        printf("Kamran Akmal");
}
```

Muhammad Ali
2025F-BIT-023
Shahid Afridi
C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 13848) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console.

2. Write a program that takes a number as input from user and checks whether the Number is even or odd.

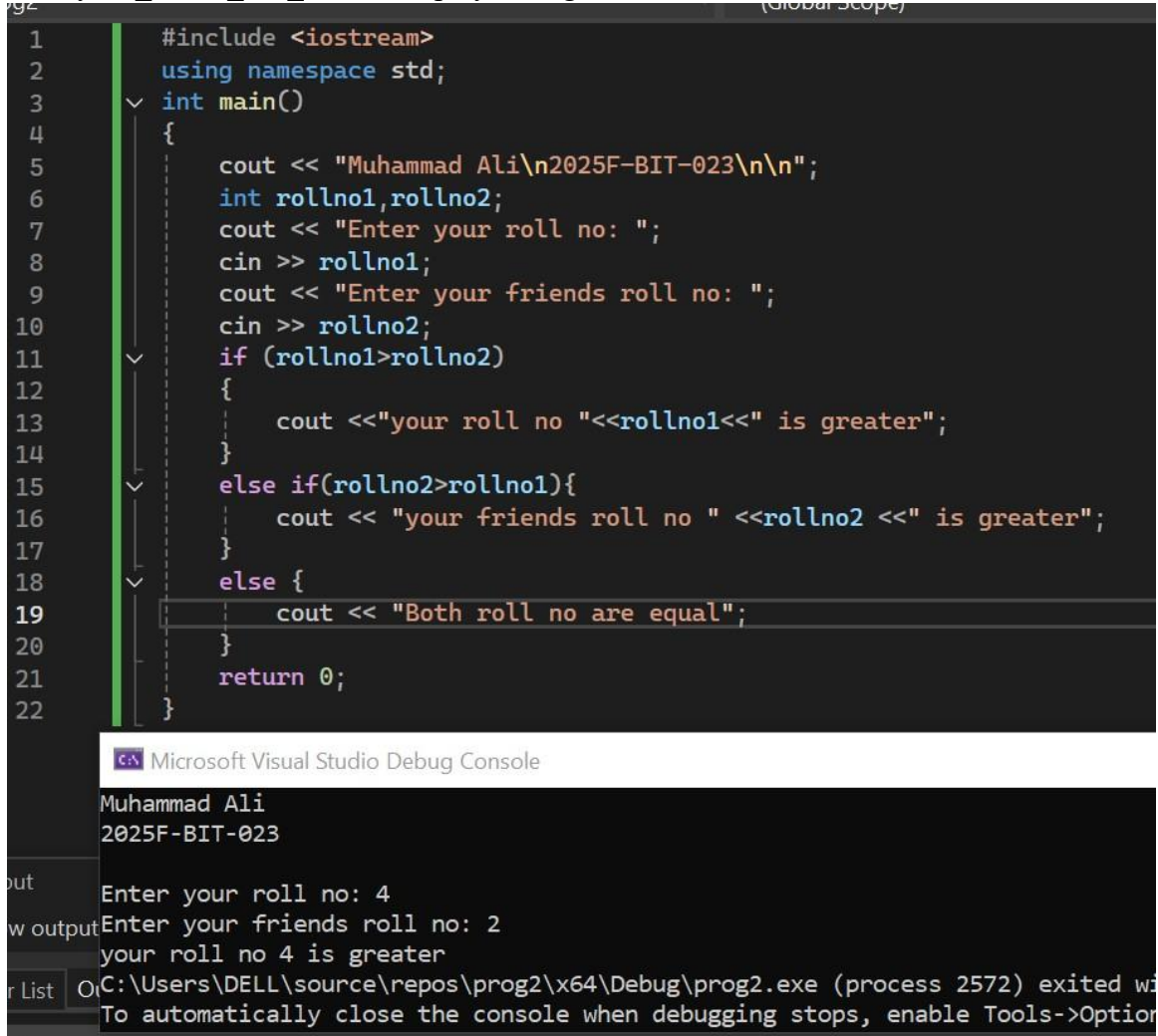
Using if-else:

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali\n2025F-BIT-023\n\n";
6      int a;
7      cout << "Enter number: ";
8      cin >> a;
9      if (a % 2 == 0)
10     {
11         cout << "The number is even";
12     }
13     else {
14         cout << "The number is odd";
15     }
16     return 0;
17 }
```

Microsoft Visual Studio Debug Console

```
Muhammad Ali
2025F-BIT-023
Enter number: 2
The number is even
C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 14568) exited with code 0 (0x0).
```

3. Write a program that declares and initializes two numbers with your_roll_no and your_friend_roll_no and displays the greater of the two.



```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali\n2025F-BIT-023\n\n";
6      int rollno1, rollno2;
7      cout << "Enter your roll no: ";
8      cin >> rollno1;
9      cout << "Enter your friends roll no: ";
10     cin >> rollno2;
11     if (rollno1 > rollno2)
12     {
13         cout << "your roll no " << rollno1 << " is greater";
14     }
15     else if (rollno2 > rollno1) {
16         cout << "your friends roll no " << rollno2 << " is greater";
17     }
18     else {
19         cout << "Both roll no are equal";
20     }
21     return 0;
22 }
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023

Enter your roll no: 4
Enter your friends roll no: 2
your roll no 4 is greater

C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 2572) exited with return code 0
To automatically close the console when debugging stops, enable Tools->Options->Debug Console->Close on Exit

4. Write a program to read the age of a candidate and determine whether it is eligible for casting his/her own vote.

```
#include <iostream>
using namespace std;
int main()
{
    cout << "Muhammad Ali\n2025F-BIT-023\n\n";
    int age;
    cout << "Enter your age: ";
    cin >> age;

    if (age>=18)
    {
        cout <<"you are eligible";
    }
    else {
        cout << "you are not eligible";
    }
    return 0;
}
```

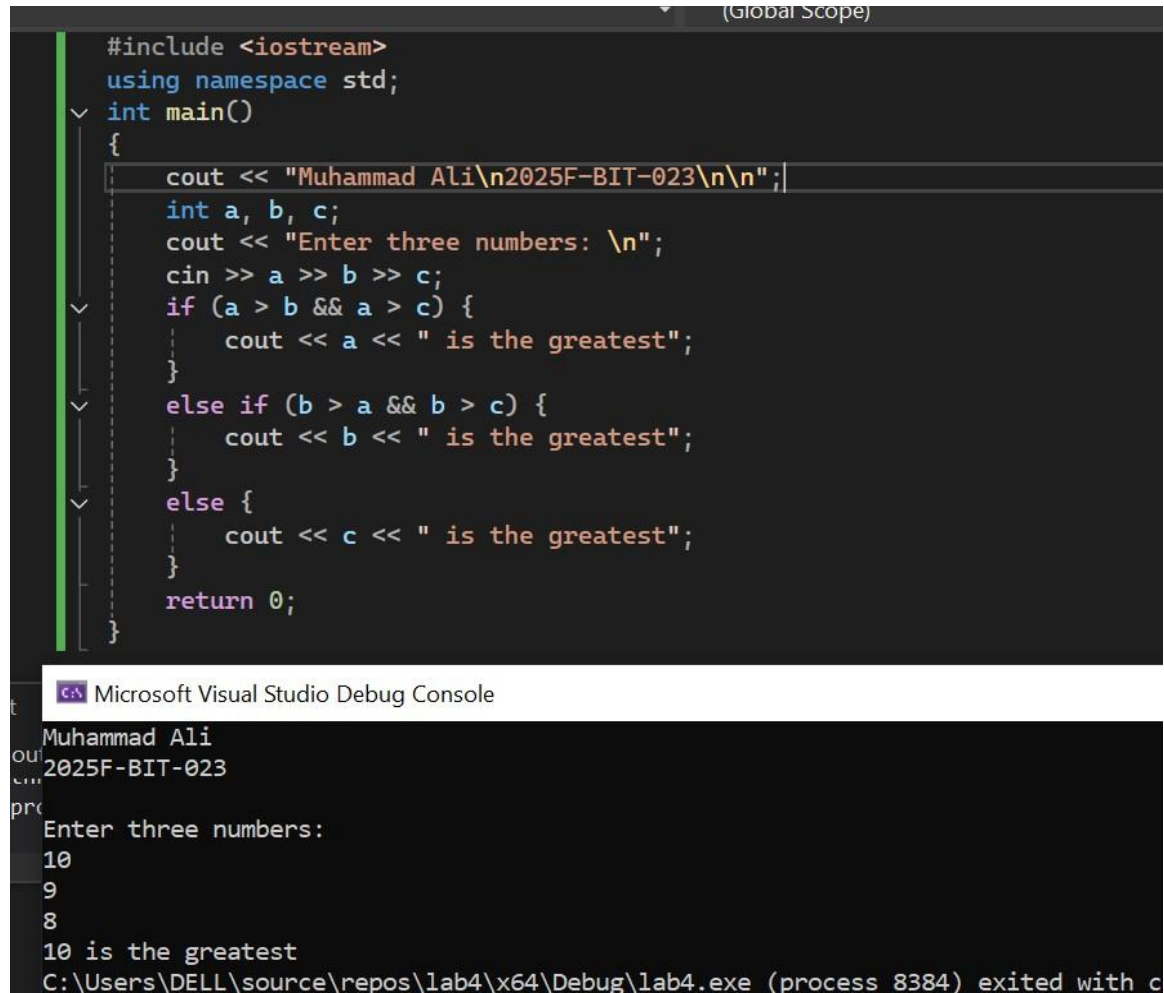
Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023

Enter your age: 18
you are eligible

C:\Users\DELL\source\repos\prog2\x64\Debug\prog2.exe (process 10164) exited with cod

5. Write a program to find the maximum between three numbers by using the if-else statement.



```
#include <iostream>
using namespace std;
int main()
{
    cout << "Muhammad Ali\n2025F-BIT-023\n\n";
    int a, b, c;
    cout << "Enter three numbers: \n";
    cin >> a >> b >> c;
    if (a > b && a > c) {
        cout << a << " is the greatest";
    }
    else if (b > a && b > c) {
        cout << b << " is the greatest";
    }
    else {
        cout << c << " is the greatest";
    }
    return 0;
}
```

Microsoft Visual Studio Debug Console

```
Muhammad Ali
2025F-BIT-023

Enter three numbers:
10
9
8
10 is the greatest
C:\Users\DELL\source\repos\lab4\x64\Debug\lab4.exe (process 8384) exited with c
```

6. Write a program to take a value from the user as input marks of five subjects Physics, Chemistry, Biology, Mathematics, and Computer. Calculate percentage and grade according to the following:

Percentage $\geq 90\%$: Grade A,

Percentage $\geq 80\%$: Grade B,

Percentage $\geq 70\%$: Grade C,

Percentage $\geq 60\%$: Grade D,

Percentage $\geq 40\%$: Grade E,

Percentage $< 40\%$: Grade F,

Write this program with the help of if-else statement

```

1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Muhammad Ali\n2025F-BIT-023\n\n";
6      float phy,chem,bio,maths,comp;
7      cout << "Enter marks of physics:";
8      cin >> phy;
9      cout << "Enter marks of chemistry:";
10     cin >> chem;
11     cout << "Enter marks of biology:";
12     cin >> bio;
13     cout << "Enter marks of maths:";
14     cin >> maths;
15     cout << "Enter marks of computer:";
16     cin >> comp;
17     float sum = phy + chem + bio + maths + comp;
18     float total_marks = 500;
19     float per = sum/total_marks*100;
20     cout << "obtained marks: "<<sum << "\nTotal marks: " << total_marks << "\nPercentage: " << per << "\n";
21     if (per >= 90) {
22         cout << "Grade A";
23     }
24     else if (per >= 80) {
25         cout << "Grade B";
26     }
27     else if (per >= 70) {
28         cout << "Grade C";
29     }
30     else if (per >= 60) {
31         cout << "Grade D";
32     }
33     else if (per >= 40) {
34         cout << "Grade E";
35     }
36     else if (per < 40) {
37         cout << "Grade F";
38     }
39     return 0;
40 }

```

Microsoft Visual Studio Debug Console

```

Muhammad Ali
2025F-BIT-023
Enter marks of physics:50
Enter marks of chemistry:60
Enter marks of biology:70
Enter marks of maths:80
Enter marks of computer:90
obtained marks: 350
Total marks: 500
Percentage: 70
Grade C
C:\Users\DELL\source\repos\lab4\x64\Debug\lab4.exe (process 20832) exited with code 0 (0x0)

```